App State & Performance Changes

Customer app (**app/**) and one backend allow-list entry

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1. Summary

This report covers two pieces of work on the customer app (**app/**), in the order they were done.

Part A - UI states. Ten states (empty, loading, error, no internet, slow network, no search result, permission denied, session expired, form validation, success) were audited screen by screen. Gaps were fixed, mainly places where a failed request looked like an empty result.

Part B - Performance and reliability. Seven items from the Play Console checklist were reviewed; six needed code changes.

No route, API contract or existing flow was changed. All new text exists in English, Telugu and Hindi.

Verification status. Typecheck passes (app and backend). The full jest suite passes: 20 suites, 150 tests (18 suites / 142 tests before Part B). The app was **not** run on a device, so nothing here has been checked on screen and no performance gain is measured.

State	Before	After
Empty	Present	Present; added meat-centre 'no results'
Loading	Present	Present
Error	Crash screen only	Shared retry view on 11 screens
No internet	Present (banner)	Present; pauses when backgrounded
Slow network	Missing	Toast after 6 s
No search result	Partial	All searches covered
Permission denied	Silent or bare alert	Open Settings alert + notifications banner
Session expired	Silent redirect	Redirect with toast
Form validation	Login accepted junk	Strict phone/email checks
Success	Payment only	Address saved, order scheduled toasts

2. Part A: UI state changes

2.1 Slow network

A request still pending after **6 s** raises a bottom toast ("Slow network - this is taking longer than usual"). It shows for 3 s, then stays quiet for 30 s so a bad connection does not spam. File uploads are excluded. **Files:** utils/api/slow-network.ts (new), components/SlowNetworkToast.tsx (new), utils/api/custom-fetch.ts, app/_layout.tsx.

2.2 Session expired

A 401 already signed the user out and redirected to login, with no explanation. The auth store now raises a *sessionExpired* flag (also on the silent sign-out at app start) and a new component turns it into the toast "Your session expired. Please sign in again." once. A normal logout shows nothing; a new sign-in clears the flag. **Files:** contexts/authStore.ts, components/SessionExpiredNotice.tsx (new), app/_layout.tsx.

2.3 Error state with retry

New shared **components/ui/ErrorState.tsx** ("Couldn't load this" + Try again). Where a failed request used to show an empty message, the error view is now shown when nothing is loaded. If data is already on screen it stays, with no error view. Only first-page failures count, not "load more".

Screen	Wrongly showed before	Now
Home	'Services unavailable' / 'No riders'	Error view (loadError in homeStore)
Orders	'No orders yet'	Error view
Offers	Own error view	Switched to shared component
Meat centres	Blank list	Error view
Notifications	'Nothing here yet'	Error view
Favourites (both tabs)	'No favourites yet'	Error view per tab
149 store	'Nothing at 149 near you'	Error view
Saved addresses	'No saved addresses'	Error view; 'use current location' stays
Restaurant menu	'Menu not available yet'	Error view; loader now checks response
Home search overlay	'No results found'	Error view (no retry button)

The restaurant menu loader also crashed on an error body (it called .map on an object); it now validates the response and supports retry.

2.4 Permission denied

- New **utils/permissions.ts**: *requestLocationPermission()* shows an alert with **Open Settings** when a user-tapped action is denied.
- Applied to 8 call sites: saved addresses, map picker, helper task pickup, ride location, ride pickup recenter, add address, delivery entry recenter, package delivery search. Add address, delivery entry and package delivery were silent before.
- Screens that ask for location on load keep their silent fallback (an unrequested popup would be worse).
- Home 'Use my current location' did nothing when permission was blocked; it now shows the alert.
- Home kept a saved *gps_permission_denied* flag forever; it now re-checks the OS status and clears it if location was enabled in Settings.
- New **NotificationsPermissionBanner** on the Notifications screen when the OS has notifications off, with an Open Settings button. It re-checks on return to foreground and registers for push if enabled. Hidden in Expo Go or if status cannot be read.

2.5 Search states

Restaurants, dishes, menu dishes, ride places and package places already had loading / no-results / error views. Gaps closed:

- **Meat centres:** 'No results found' view for an empty search or filter (and the error view from 2.3).
- **Add address and Helper task place search** had no loading, empty or error line, no debounce and no stale-response guard. New **features/delivery/placeSearch.ts** adds a 300 ms debounce, ignores out-of-date responses, and drives a status line (Searching / No matching places / Couldn't load) via the new **PlaceSearchMessage**. Minimum query length is now 2 characters on Add address.
- Ride search error text was hardcoded English; now translated. Home search headings 'RECENTLY SEARCHED' and 'RECOMMENDED' are now translated.

2.6 Form validation

Login 'Continue with OTP' accepted anything containing 10 digits, so an email like user1234567890@mail.com sent a code to a number nobody typed. New **features/auth/login.validation.ts** requires a real 10-digit mobile (spaces, dashes, +91 or leading 0 allowed; letters or '@' rejected), and password sign-in checks for a valid phone or email first. 4 unit tests added. Signup, personal details, OTP and add address were already correct and left alone.

2.7 Success states

Saving an address closed the screen silently, and a scheduled food order dropped the user on Orders with no feedback. Both now show a toast ("Address saved"; "Order scheduled. We'll confirm once the restaurant accepts."). A new *showToast()* in components/ui/Toast.tsx lets non-component code raise toasts. Online payment already had its full-screen success/failure animation.

2.8 Not covered by Part A

- Live ride flows (tracking, finding driver, ride confirmation), profile, language settings and checkout were not audited for these states.
- Support flow still has a few hardcoded English alerts ('Error', 'Please try again.', 'Failed to reopen case').
- No-internet detection is a 15 s ping, not an instant OS event, so the banner can appear up to ~15 s late.

#	Checklist item	Outcome	Type of change
1	Optimize app startup time	Push setup deferred until after first interactions	Code
2	Reduce battery, CPU, memory	11 pollers pause while app is backgrounded	Code
3	Compress and cache images	Cloudinary resized/compressed thumbnails in 9 files	Code
4	Avoid blocking main thread	Reviewed; no change needed	None
5	Slow networks / connection loss	30 s request timeout; 2 silent failures now toast	Code
6	Crashes, ANRs, background work	App error reporting added (JS + render crashes)	Code + backend
7	Low storage / low memory	Login can no longer fail because storage is full	Code

3. Part B: Performance and reliability changes

3.1 Startup time

Push-notification registration (creating Android channels, fetching the Expo push token) used to run immediately after sign-in, competing with the launch animation and first screen for the JavaScript thread. It now runs inside *InteractionManager.runAfterInteractions* and is cancelled on cleanup.

The ADIOS letter animation was deliberately left untouched because it is part of the existing flow. **File:** utils/usePushNotifications.ts

3.2 Battery, CPU and memory

New helper **utils/foregroundInterval.ts** exposes *setForegroundInterval*, a drop-in for *setInterval* that skips ticks while *AppState* is not active. Because it returns a normal interval id, existing *clearInterval* cleanups work unchanged. Polling a server for a screen nobody can see keeps the radio and CPU awake; this removes that cost.

Poller	Interval	File
Offline banner ping	15 s	components/OfflineBanner.tsx
Support tickets refresh	4 s	features/support/useSupportChatHandleCreateTicket.tsx
Package trip driver refresh	config	features/package-delivery/usePackageDeliveryTrip.ts

Poller	Interval	File
Outlet ordering state	30 s	features/food/useOutletOrderingState.ts
Finding driver: order status	2 s	features/ride/useFindingDriverInsets.effects.ts
Finding driver: online drivers	10 s	features/ride/useFindingDriverShowCancelSheet.tsx
Ride confirmation: nearby drivers	12 s	features/ride/useRideConfirmationMapFit.tsx
Ride searching: online drivers	10 s	features/ride/useRideSearchingCancelReasonVisible.tsx
Tracking: order details	7 s	features/ride/useTrackingPickupStop.tsx
Helper task: status	3 s	features/delivery/useHelperTaskLiveStatus.tsx
Helper task: poll	5 s	features/delivery/useHelperTaskLiveStatus.tsx

The offline banner additionally re-checks once when the app returns to the foreground, so it is not stale after being backgrounded. Socket connections are unchanged: they remain the primary live-update path.

3.3 Images

New helper **utils/imageUrl.ts** (*optimizedImageUrl*) inserts a Cloudinary transformation (*f_auto,q_auto,w_<width>,c_limit*) so list thumbnails download at roughly the size they are shown (width x device pixel ratio, capped at 1080 px) in WebP/AVIF where supported. It never upscales, leaves URLs that already carry a transformation alone, and returns non-Cloudinary or empty values untouched. *expo-image* was already providing disk and memory caching.

Applied to: RestaurantListItem, Store149Card, Store149Sheet, MenuBody, DishSearchResultItem, FavoriteCard, ItemFavoriteCard, OfferRestaurantCard, CartContents. Hero images were not changed.

3.4 Main thread

Reviewed, no code change. React Compiler is enabled (app.config.js), animations use Reanimated on the UI thread, and Home uses FlashList. Recommend profiling Home filtering and sorting if lists grow beyond the current 20-per-page.

3.5 Slow networks and connection loss

- **Request timeout.** *customFetch* now aborts a request after 30 s and throws a translated "request timed out" message. Previously a request on a dead connection could leave a screen on its spinner indefinitely. File uploads and calls that supply their own abort signal are excluded. (utils/api/custom-fetch.ts)
- **Support cases list:** a failed first load now shows a toast instead of failing silently. The 4 s background refresh stays silent to avoid repeated toasts on weak signal.
- **Restaurant details:** a failed vendor lookup now shows a toast; the page still renders what the previous screen passed along.
- New strings added in English, Telugu and Hindi.

3.6 Crashes, ANRs and background work

There was no crash reporting; a crash on a customer's phone left no trace. New **utils/crashReporting.ts** sends an *app_error* event (kind, fatal flag, message, first stack frames, each capped at 200 characters) through the existing analytics pipeline to Firebase Analytics and the backend Live Activity store.

- Installed once at startup via a global JS error handler; the previous handler (red box / crash) still runs.
- The ErrorBoundary now reports render crashes before showing the existing fallback screen.
- Identical errors are de-duplicated for 30 s to avoid flooding from a render loop.
- A fatal error triggers an immediate upload through the new *flushAnalytics()*.

- **Backend:** *app_error* added to ALLOWED_EVENTS in backend/src/modules/analytics/analytics.service.ts. **Without deploying this the server drops the events.**

3.7 Low storage and memory

- Saving the session after login (token and user) is now wrapped so a full or failing disk cannot turn a successful login into a "Login failed" alert. The worst case is signing in again after the app is closed. (contexts/auth.credentials.ts)
- *setToken* and the startup-ad date write no longer produce unhandled promise rejections.
- Cart, address and theme writes were already guarded.

4. Files changed

New files (Part A)	Purpose
components/ui/ErrorState.tsx	Shared error view with retry
components/SlowNetworkToast.tsx, utils/api/slow-network.ts	Slow network toast
components/SessionExpiredNotice.tsx	Session expired toast
utils/permissions.ts	Open Settings alert, location request
features/profile/components/NotificationsPermissionBanner.tsx	Notifications-off banner
features/delivery/placeSearch.ts, components/PlaceSearchMessage.tsx	Place search status, debounce
features/auth/login.validation.ts (+ test)	Login phone/email checks

Part A also edited about 40 existing files (screens, hooks, stores and the three locale files) to wire these in.

New files (Part B)	Purpose
utils/foregroundInterval.ts	Interval that pauses in background
utils/imageUrl.ts	Cloudinary resize/compress URL helper
utils/crashReporting.ts	JS and render error reporting
utils/__tests__/imageUrl.test.ts	4 tests
utils/__tests__/crashReporting.test.ts	4 tests

Modified (core)	Change
utils/api/custom-fetch.ts	30 s timeout
utils/analytics.ts	Exported flushAnalytics()
utils/usePushNotifications.ts	Deferred registration
components/ErrorBoundary.tsx	Reports render errors
app/_layout.tsx	Installs crash reporting
contexts/auth.credentials.ts, authStore.ts	Safe session persistence
features/home/useHomeSearchAndBanners.tsx	Guarded startup-ad write
features/support/useSupportChatInsets.tsx	Failure toast
features/food/useRestaurantDetails.tsx	Failure toast
locales/en|te|hi/common.json	New strings
backend/.../analytics.service.ts	app_error allowed

Plus the 11 polling files in section 3.2 and the 9 image files in section 3.3 (one-line edits each).

5. Known limits

- **Not a full crash service.** No symbolicated stack traces; native crashes and ANRs are not captured. For those, add Firebase Crashlytics or Sentry (requires a new native build).
- Unhandled promise rejections are not captured.
- Startup time, battery and memory are **not measured**; the changes follow known causes, not profiling results.
- The 30 s timeout is a single global value. Slow but valid calls (large payloads on 2G) could hit it; adjust REQUEST_TIMEOUT_MS if testing shows that.
- Support ticket polling, ride and helper polling resume at the next tick after returning to foreground, so a screen may update up to one interval later than before in that case.

6. Deployment and rollout

- **Backend first:** deploy the analytics allow-list change, otherwise app_error events are discarded.
- App changes ship with the next build; no native module was added, so an OTA update is sufficient.
- No database migration, no API change, no environment variable change.

7. How to verify

Area	Check	Expected
Polling	Background the app on finding-driver or tracking; watch network log	Requests stop, resume on return
Timeout	Proxy that never answers; open Home	Fails after about 30 s with timeout message
Crash events	After backend deploy, force an error	app_error appears on admin Live Activity
Images	Inspect a thumbnail request	URL contains f_auto,q_auto,w_...
Low storage	Emulator with nearly full disk; log in	Login succeeds, no error alert
Startup	adb shell am start -W (cold start) before and after	Record numbers; target under 2 s to first frame
Support toast	Airplane mode, open support cases	Toast on first load only
Error views	Airplane mode, kill app, reopen; open each screen in 2.3, then restore network and tap Try again	'Couldn't load this', then recovers
Slow network	Throttle to very slow; open Home	Toast after ~6 s, shown 3 s
Session expired	Invalidate token, open any screen	Lands on login once with toast
Permissions	Deny location (twice on Android), tap a 'use current location' button	Alert with Open Settings
Notifications banner	Turn notifications off for the app; open Notifications	Banner shows; hides after enabling
Search	Search 'zzzz' in meat centres, add address, helper task	'No results' / 'No matching places'
Login validation	Try abc, 12345, user1234567890@mail.com	Refused with message
Success toasts	Save an address; place a scheduled food order	Toast shown

Suggested next step: measure cold start, idle battery and memory on a low-end device (2 GB RAM) to confirm these changes and decide whether Crashlytics or Sentry is worth adding.